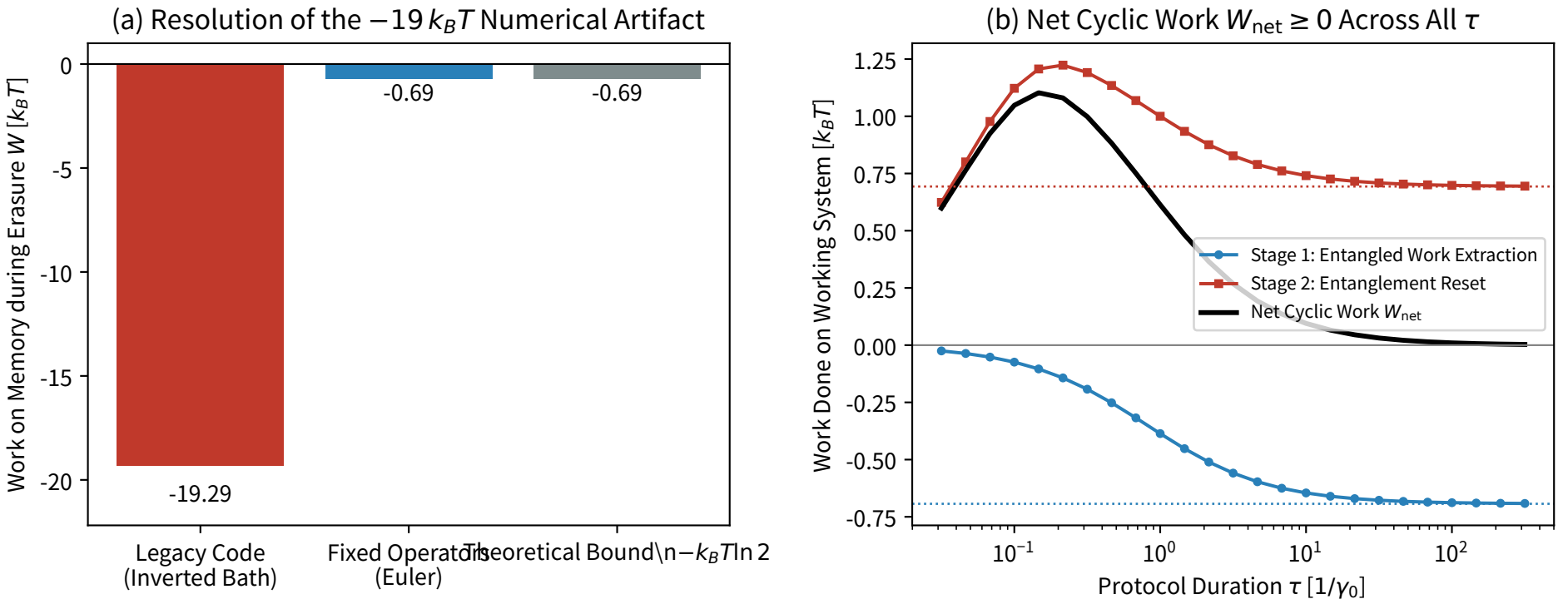

(a) Resolution of the $-19\,k_BT$ Numerical Artifact
Work on Memory during Erasure W $[k_BT]$
0
-5
-10
-15
-20
-19.29
-0.69
-0.69
Legacy Code
(Inverted Bath)
Fixed Operator
(Euler)
Theoretical Bound\n$-k_BT\ln 2$
(b) Net Cyclic Work $W_{\text{net}} \geq 0$ Across All τ
Work Done on Working System $[k_BT]$
1.25
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
10^{-1}
10^{0}
10^{1}
10^{2}
Protocol Duration τ $[1/\gamma_0]$
Stage 1: Entangled Work Extraction
Stage 2: Entanglement Reset
Net Cyclic Work W_{net}